# Robotics – Lab assignment 1: Golf swing in MuJoCo

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## 1. Task 1 – MuJoCo model of player + club

**Structure.** A kinematic tree rooted at a fixed `torso` body at $(0, 0, 1.255\,\text{m})$. The lower body is rigid and welded to the base; the actuated chain is torso → waist → upper torso → shoulders → upper arms → lower arms → hands → club. The ball is a separate free body.

**Degrees of freedom (11 actuated).** Torso yaw (1), each shoulder (3), each elbow (1), right wrist cock and deviation (2). The ball adds a `freejoint` (6 DOF), giving $n_v = 17$ velocity and $n_q = 18$ position coordinates (the extra entry is the ball quaternion); ranges in Table 1.

**Reference frames.** Each body's `pos`/`quat` is defined *relative to its parent*, so every link carries its own frame and all descendant frames rotate with it – this is what makes the chain kinematic. The world frame is $z$-up at ground level, $g = -9.81\,\text{m}\,\text{s}^{-2}$. Three *sites* are used for measurement: `shared_grip` and `left_grip` (hand constraint) and `club_head` (impact speed, via its Jacobian).

**Mass and inertia.** Each geom carries an explicit mass; MuJoCo derives the centre of mass and the inertia tensor from the geom shape assuming *uniform density*. Player total is exactly 65 kg (leg 11 kg each, foot 1 kg, torso+waist+pelvis 30 kg, arm 3 kg, head 5 kg). The club mesh uses `density="1000"`, giving 0.382 kg over the scaled volume – in the range of a real club (0.31–0.42 kg). The ball is 0.046 kg, radius 21.3 mm, matching the regulation ball (45.93 g, 21.33 mm).

**Friction at each DOF.** Every hinge has `damping`=2 N m s rad$^{-1}$ (viscous joint friction) and `armature`=0.01 (rotor inertia). Body capsules use `condim="1"` (normal-only contact); the club-head/ball pair has its own `condim="3"` contact, tangential friction 0.3, with a short deliberately underdamped response (`solref="0.003 0.35"`).

**Actuation.** All 11 DOF use `motor` actuators – direct torque sources in N m: torso ±250, shoulder DOF ±180, elbows and wrists ±90. The ball has no actuator; it moves only from contact and gravity.

**Control.** A reference is built offline from five key poses (torso angle, grip position in the upper-torso frame, club phase), solved by inverse kinematics and splined to give $q_d, \dot{q}_d, \ddot{q}_d$. At runtime a PD law with inverse-dynamics compensation produces the torque, so the arms move *only* from applied torques – never by writing joint angles:

$$\tau = M(q)\,\ddot{q}_{\text{cmd}} + C(q, \dot{q}), \qquad \ddot{q}_{\text{cmd}} = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + \ddot{q}_d,$$

with gains from a damping ratio of 1 and a 0.30 s settling time, integrated at $\Delta t = 0.5$ ms (2 kHz, Euler).

**Files.** `human.xml` (commented), `golf_club.stl` (visual) and `golf_club_head.stl` (separate convex collider – the hull of the whole club would otherwise form an invisible wedge between shaft and head), plus `swing_trajectory.py`, `main.py`, `disturbance_sweep.py`, `analysis.py`, `export_visuals.py`.

### Simplifications and their impact on realism

- *Lower body rigid.* No weight transfer, balance loss or foot slip. A real swing draws much of its club-head speed from legs and hips, so our 9.4 m s$^{-1}$ is far below a real player; the model captures the arm/torso contribution only.
- *Hands welded* by an `equality/connect` constraint: no grip slip or release, so the grip can never fail – optimistic.
- *Capsules of uniform density.* Limb mass distribution is idealised, so segment inertias – and the torques they demand – are approximate.
- *Rigid club.* Real shafts bend and release energy at impact ("kick"), adding speed we cannot reproduce.
- *No aerodynamics.* The ball flies as a pure parabola – no drag, no backspin lift – so carry distance is not comparable to a real shot.
- *Ideal actuators.* No electrical dynamics, PWM, backlash or delay; the commanded N m is applied exactly, within limits.
- *Contact tuned, not measured.* `solref`/`solimp` were chosen for a stable strike, not fitted to a measured coefficient of restitution.
- *Reference poses hand-designed*, not motion capture: the swing approximates Figure 1 of the guide rather than a recorded human swing.

## 2. Task 2 – Tests designed and run

All tests run headless from Python (`main.py`, `disturbance_sweep.py`, `analysis.py`); the model also loads directly in the Simulate app.

1. **Nominal swing (baseline).** Full 3.5 s swing; records impact time, club-head speed (from the `club_head` site Jacobian), ball launch velocity, tracking error, saturation and grip error (Table 3).
2. **Disturbance rejection per DOF** (required by the guide). An external torque is added to `qfrc_applied` – *not* to `ctrl` – so the controller must actively reject it: $\tau_{\text{dist}} = A \sin^2(\pi u) \sin(2\pi f \tau)$, $f = 8$ Hz, with a $\sin^2$ window that is zero at the start and end of the swing (no step jolt). Each of the 11 joints was disturbed separately at $A \in \{30, 60, 300\}$ N m (Table 4).
3. **Controller detuning.** PD settling time swept over $\{0.3, 0.75, 5, 10, 15, 20\}$ s (Table 2): the transition from accurate tracking to total loss of the swing.
4. **Reduced-torque check.** A runtime key (`P`) scales all motor commands to 10%, including gravity compensation. The body visibly struggles to follow the reference, confirming that the motion is produced by active torques against dynamics, not by a scripted animation.
5. **Structural checks, every run.** Grip closure, joint excursions against the XML limits, and MuJoCo warning counters (zero in all runs).

## 3. Task 3 – Comment on the results

**What works well.** The controller tracks the designed swing to a maximum error of 1.03° with zero actuator saturation and no MuJoCo warnings, so the simulation is numerically clean and the motors are not being asked for impossible torques. The strike is consistent: contact at 2.253 s with a club-head speed of 9.44 m s$^{-1}$, launching the 46 g ball at 11.42 m s$^{-1}$. The hand constraint holds to 0.013 mm nominally and under 0.6 mm even under the most violent disturbance. Disturbance rejection at 30 N m is strong: every joint stays below 9° of added error with no saturation, so the model tolerates realistic tremor. Most convincingly, the sensitivity ranking the simulation produces is the physically correct one: the lead (left) arm dominates – `left_elbow`, `left_arm_lift` and `left_shoulder1` show the largest errors at every amplitude – which is exactly what golf biomechanics predicts, since the straight lead arm sets the radius and plane of the swing arc. The torso, with the largest motor and the shortest lever to the club, is the least affected. This pattern was not designed in; it emerged from the dynamics, which is good evidence the model behaves like a mechanism rather than an animation.

**Caveats.** (i) *Ball distance is a poor damage metric*: disturbances can add energy or change launch direction, sending the ball further while tracking degrades. Tracking error, saturation and whether contact occurred are the trustworthy indicators; we report distance but do not rank by it. (ii) *Saturation, not tracking error, predicts catastrophic failure*: `left_arm_lift` reaches 58.5° of error yet still strikes the ball cleanly (0.9% saturation, 11.3 m), while the two total misses at 300 N m (`right_shoulder1`, `left_elbow`) are the most heavily saturated cases (17.4% and 34.2%). At this level the disturbance exceeds every actuator limit, so no choice of gains could recover. (iii) The disturbance is deterministic (sin, not noise), so results are perfectly reproducible but not a true stochastic tremor; band-limited noise would be more representative of a genuinely shaky joint. (iv) Equal amplitude across joints is not a fair comparison: 60 N m is 67% of an elbow's capacity but only 24% of the torso's, so part of the ranking reflects motor sizing rather than kinematic role. A normalised sweep (fixed fraction of each actuator's limit) would separate the two effects. (v) *The fixed lower body is the largest single simplification.* In a real swing the kinetic chain starts at the feet: ground reaction forces accelerate the hips, which lead the torso, which leads the arms. Our model captures only the last two links of this chain, which is why the club-head speed (9.4 m s$^{-1}$) is roughly a quarter of a real amateur swing ($\sim$35 m s$^{-1}$). Absolute distances are therefore not comparable to real golf; the model is valid for *relative* comparisons – which joint matters, how gains affect tracking – not for predicting shot distance. Freeing the lower body would require balance control and foot-ground contact modelling, substantially increasing the model's complexity.

## Appendix – Data

Table 1: Degrees of freedom and actuation limits.

| Joint group | DOF | Angular range | Max torque |
|---|---|---|---|
| Torso (yaw) | 1 | $-90^\circ$ to $90^\circ$ | $\pm 250$ N m |
| Right shoulder | 3 | $-170^\circ$ to $90^\circ$; $\pm 170^\circ$; $\pm 170^\circ$ | $\pm 180$ N m |
| Right elbow | 1 | $-32.6^\circ$ to $110^\circ$ | $\pm 90$ N m |
| Right wrist | 2 | $-150^\circ$ to $150^\circ$; $-120^\circ$ to $100^\circ$ | $\pm 90$ N m |
| Left shoulder | 3 | $-170^\circ$ to $90^\circ$; $\pm 170^\circ$; $\pm 170^\circ$ | $\pm 180$ N m |
| Left elbow | 1 | $-32.6^\circ$ to $110^\circ$ | $\pm 90$ N m |
| **Player total** | **11** | | |
| Golf ball (freejoint) | 6 | – | none |

Table 2: Controller detuning: PD settling time vs. tracking and strike.

| Settling time [s] | 0.30 | 0.75 | 5 | 10 | 15 |
|---|---|---|---|---|---|
| Club-head speed [m/s] | 9.44 | 9.44 | 9.39 | 9.32 | 5.29 |
| Ball displacement [m] | 11.05 | 10.95 | 11.31 | 11.17 | 6.67 |
| Max tracking error [deg] | 1.03 | 2.02 | 9.11 | 14.49 | 82.79 |
| Saturated steps [%] | 0.00 | 0.00 | 0.00 | 0.00 | 0.86 |

At 20 s the controller fails completely: tracking error $238.5^\circ$, the club strikes the ground before reaching the ball and no contact occurs ($d = 0$). Gains were therefore kept at the 0.30 s setting.

Table 3: Baseline (undisturbed) swing, 3.5 s, $\Delta t = 0.5$ ms.

| | | | |
|---|---|---|---|
| Impact time | 2.253 s | Peak ball speed | 11.42 m s$^{-1}$ |
| Club-head speed at impact | 9.44 m s$^{-1}$ | Ball horizontal displacement | 11.05 m |
| Ball launch velocity $(x, y, z)$ | (2.02, 11.13, 1.55) m/s | Max ball height | 0.148 m |
| Max tracking error | $1.03^\circ$ | Max grip error | 0.013 mm |
| Saturated steps | 0.00 % | MuJoCo warnings | none |

Table 4: Disturbance sweep: ball distance $d$ [m], max tracking error $e$ [deg], saturated steps $s$ [%]. Baseline $d = 11.05$, $e = 1.03$, $s = 0$. Ordered by sensitivity at 60 N m.

| | $A = 30$ N m | | | $A = 60$ N m | | | $A = 300$ N m | | |
|---|---|---|---|---|---|---|---|---|---|
| Joint | $d$ | $e$ | $s$ | $d$ | $e$ | $s$ | $d$ | $e$ | $s$ |
| left_elbow | 10.90 | 8.5 | 0.0 | 6.66 | 18.1 | 4.6 | **miss** | 80.1 | 34.2 |
| left_arm_lift | 10.93 | 5.6 | 0.0 | 11.23 | 11.3 | 0.0 | 11.34 | 58.5 | 0.9 |
| left_shoulder1 | 11.05 | 3.8 | 0.0 | 11.66 | 7.6 | 0.0 | 12.97 | 36.0 | 10.1 |
| right_shoulder1 | 11.44 | 1.0 | 0.0 | 7.37 | 4.5 | 0.7 | **miss** | 37.7 | 17.4 |
| left_shoulder2 | 12.21 | 2.2 | 0.0 | 10.52 | 4.3 | 0.0 | 9.93 | 23.1 | 3.1 |
| right_wrist_cock | 12.15 | 1.9 | 0.0 | 11.38 | 3.7 | 0.0 | 12.82 | 95.1 | 36.2 |
| right_shoulder2 | 11.46 | 1.2 | 0.0 | 11.79 | 2.5 | 0.0 | 11.06 | 21.1 | 5.2 |
| right_elbow | 12.45 | 1.2 | 0.0 | 11.98 | 1.9 | 0.0 | 9.94 | 59.2 | 24.2 |
| right_wrist_deviation | 12.41 | 1.2 | 0.0 | 11.92 | 1.7 | 0.3 | 7.55 | 49.9 | 29.3 |
| torso_turn | 12.47 | 1.4 | 0.0 | 12.11 | 1.6 | 0.0 | 12.35 | 5.0 | 2.6 |
| right_arm_lift | 11.04 | 0.9 | 0.0 | 12.57 | 1.0 | 0.0 | 7.39 | 17.4 | 11.4 |